

Scalability & Future Plan

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the Flood Prediction System Using Machine Learning can be scaled to support larger datasets, more users, and additional features in future releases. It also presents a roadmap for enhancing the system beyond its current implementation.

Current System Limitations

S.N o	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application is hosted on a local Flask server and supports a limited number of concurrent users.	Reduced scalability for public deployment.	High
2	Uses a CSV dataset instead of a centralized database.	Limited data storage and management capabilities.	Medium
3	Prediction is based on historical weather data only and does not include live weather information.	May reduce prediction accuracy for real-time events.	High

Scalability Plan

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only	Deploy on cloud platforms (AWS, Azure, or IBM Cloud) with load balancing to handle thousands of concurrent users.
2	Data Storage	Historical data stored in CSV files	Migrate to MySQL, PostgreSQL, or MongoDB for scalable and efficient data management.
3	Performance	Single Flask server with locally trained ML model	Optimize model inference, use caching, and deploy with Gunicorn/Nginx or Docker containers for better performance.
4	Security	Basic input validation	Implement user authentication, HTTPS, API security, and role-based access control for secure access.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather and rainfall APIs (e.g., IMD/OpenWeather)	3–6 Months	Improve prediction accuracy using live environmental data.

Phase 3	Develop Android and iOS mobile applications with instant flood alerts	6–9 Months	Increase accessibility and provide timely notifications to users.
Phase 4	Deploy on cloud infrastructure with GIS-based flood mapping and IoT sensor integration	9–12 Months	Enable large-scale deployment, real-time monitoring, and smarter disaster management.

Future Vision

The long-term goal of the Flood Prediction System Using Machine Learning is to evolve into a comprehensive disaster management platform by integrating real-time weather services, cloud computing, IoT-based water level sensors, and Geographic Information System (GIS) mapping. These enhancements will improve prediction accuracy, support a larger user base, and assist governments and disaster management authorities in making faster, data-driven decisions for flood preparedness and response.